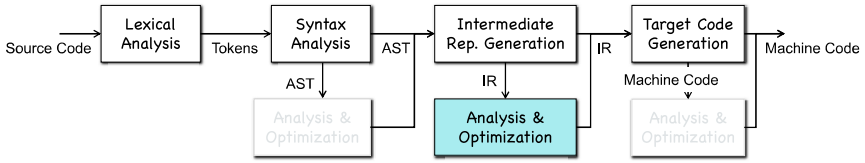
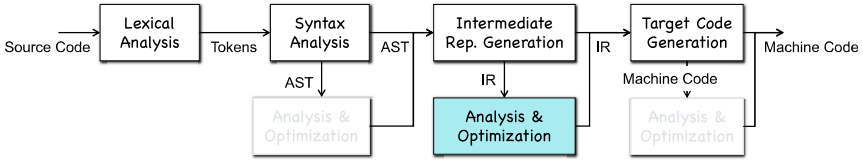
Chapter 9-1
Data Flow Analysis (DFA)

Data Flow Analysis



- IR is friendly to analysis and optimization in compilers
  - Rich source code information like types (compared to machine code)
  - Standard format, control flow graphs, etc. (compared to source code)
  - …

Data Flow Analysis



- Data flow analysis is a form of static program analysis
  - **Machine-independent compiler optimization**
  - Detection of software errors and vulnerabilities
  - …

Sources of Optimization

- Redundancy in source code

```
std::vector<int> Vec;
...
for (int K = 0; K < Vec.size();
    ...
}
```

```
struct A { int x; int y; } ;
...
A *Obj = new A;
printf("%d\n", Obj->y);
printf("%d\n", Obj->y);
```

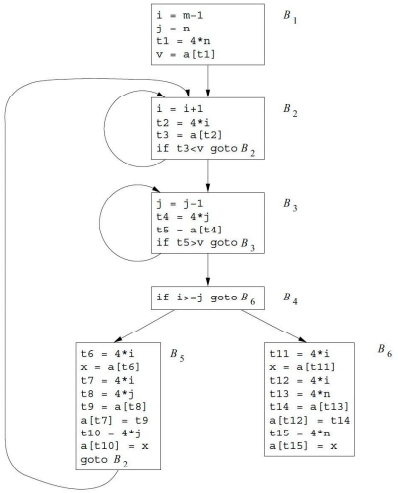
```
; Function Attrs: noinline norecurse optnone ssp uwtable mustprogress
define dso_local i32 @main() #0 {
  %1 = alloca i32, align 4
  %2 = alloca %struct.A*, align 8
  store i32 0, i32* %1, align 4
  %3 = call noalias nonnull i8* @_Znwm(i64 8) #3
  %4 = bitcast i8* %3 to %struct.A*
  store %struct.A* %4, %struct.A** %2, align 8
  %5 = load %struct.A*, %struct.A** %2, align 8
  %6 = getelementptr inbounds %struct.A, %struct.A* %5, i32 0, i32 1
  store i32 1, i32* %6, align 4
  %7 = load %struct.A*, %struct.A** %2, align 8
  %8 = getelementptr inbounds %struct.A, %struct.A* %7, i32 0, i32 1
  %9 = load i32, i32* %8, align 4
  %10 = call i32 (i8*, ...) @printf(i8* getelementptr inbounds ([4 x i8], [4 x i8]* @.str, i64 0,
  %11 = load %struct.A*, %struct.A** %2, align 8
  %12 = getelementptr inbounds %struct.A, %struct.A* %11, i32 0, i32 1
  %13 = load i32, i32* %12, align 4
  %14 = call i32 (i8*, ...) @printf(i8* getelementptr inbounds ([4 x i8], [4 x i8]* @.str, i64 0,
  ret i32 0
}
```

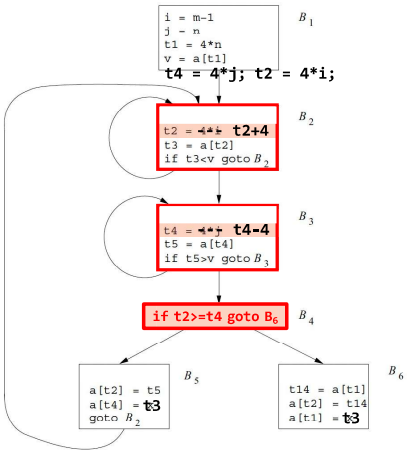

Quick Sort: A Running Example

```
void quicksort(int m, int n)
    /* recursively sorts a[m] through a[n] */
{
    int i, j;
    int v, x;
    if (n <= m) return;
    /* fragment begins here */
    i = m-1; j = n; v = a[n];
    while (1) {
        do i = i+1; while (a[i] < v);
        do j = j-1; while (a[j] > v);
        if (i >= j) break;
        x = a[i]; a[i] = a[j]; a[j] = x; /* swap a[i], a[j] */
    }
    x = a[i]; a[i] = a[n]; a[n] = x; /* swap a[i], a[n] */
    /* fragment ends here */
    quicksort(m,j); quicksort(i+1,n);
}
```


Quick Sort: A Running Example

```
(1)    i = m-1              (16)    t7 = 4*i
(2)    j = n                (17)    t8 = 4*j
(3)    t1 = 4*n             (18)    t9 = a[t8]
(4)    v = a[t1]            (19)    a[t7] = t9
(5)    i = i+1              (20)    t10 = 4*j
(6)    t2 = 4*i             (21)    a[t10] = x
(7)    t3 = a[t2]           (22)    goto (5)
(8)    if t3<v goto (5)     (23)    t11 = 4*i
(9)    j = j-1              (24)    x  = a[t11]
(10)   t4 = 4*j             (25)    t12 = 4*i
(11)   t5 = a[t4]           (26)    t13 = 4*n
(12)   if t5>v goto (9)     (27)    t14 = a[t13]
(13)   if i>=j goto (23)    (28)    a[t12] = t14
(14)   t6 = 4*i             (29)    t15 = 4*n
(15)   x = a[t6]            (30)    a[t15] = x
```





1. Local common subexpression
2. **Global common subexpression**
3. **Copy propagation**
4. **Strength reduction**
5. **Induction-variable elimination**
6. **……**

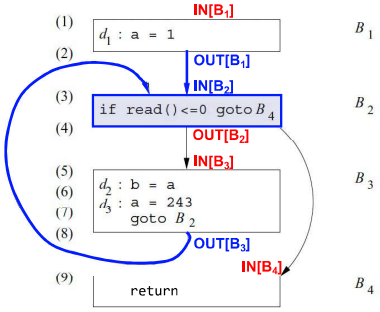
How?

PART I: DFA Scheme

Data Flow Analysis

- Describes how “data flow facts” transfer in a program, which often contains unbounded number of execution paths.
- **Data flow fact** (or value) is an abstraction of all possible concrete program behaviors, *e.g.*,
  - A variable v may have one of the concrete values { 1, 2, 4, 5, 6, … }
  - Impossible to enumerate all cases in a data flow analysis
  - **Abstraction**: $v = \text{NAC}$;
  - NAC is an abstract value of the concrete values { 1, 2, 4, 5, 6, … }

Data Flow Analysis



- IN[B]/OUT[B]
  - Set of data flow facts before and after a block
- Constraints
  - OUT[B] = f_B(IN[B])
  - IN[B] = $\wedge_{P \in preds(B)}$ OUT[P]
  - => **Data Flow Equations**

Data Flow Analysis

- To perform data flow analysis, define
  - transfer function f for each statement/block
  - merge function ∧ at the joint point of multiple paths
  - the initial set of data flow facts for each block
- Data flow analysis produces
  - IN[B]/OUT[B] that include data flow facts at a program point
  - The resulting data flow facts are always true during program execution

The Worklist Algorithm

- **ForEach** basic block B: initialize IN[B] and OUT[B]; **EndFor**
- worklist = set of all basic blocks;

- **While** (!worklist.empty()) **Do**
- B = worklist.pop();
- IN[B] = $\wedge_{P \in preds(B)}$ OUT[P]; OUT[B] = f_B(IN[B])
- **If** OUT[B] changed: worklist.push(all B's successors); **EndIf**
- **EndWhile**

Reaching Definitions

- A definition *d* may reach a program point, if there is a path from *d* to the program point such that *d* is not killed along the path.

- OUT[B] = f_B(IN[B]) = $gen_B \cup$ (IN[B] - $kill_B$)
- IN[B] = $\wedge_{P \in preds(B)}$ OUT[P] = $\cup_{P \in preds(B)}$ OUT[P]

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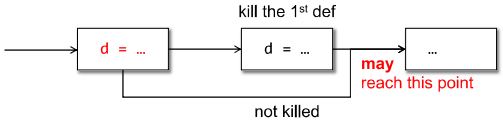
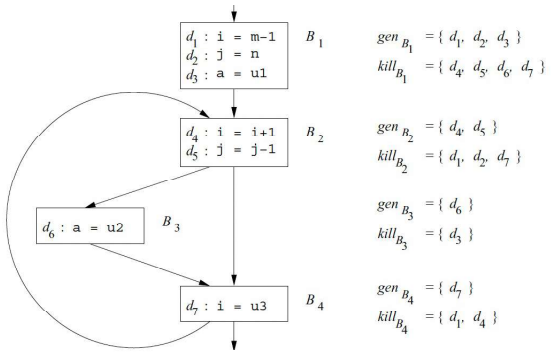


Reaching Definitions



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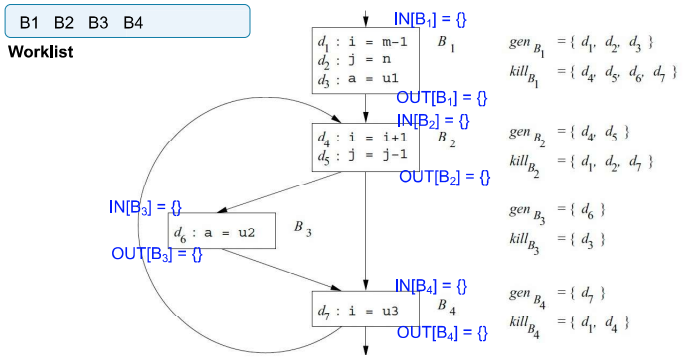
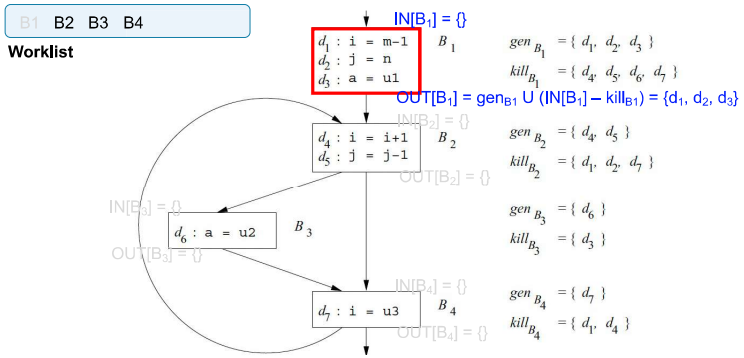


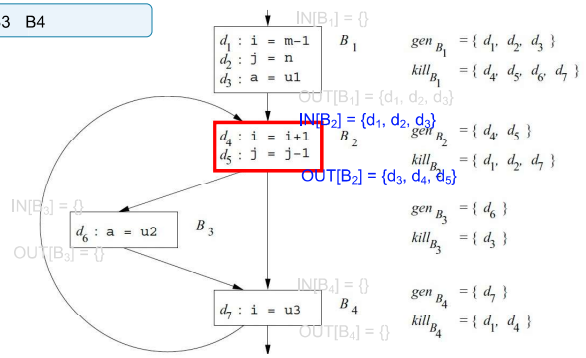
Reaching Definitions



Reaching Definitions

B1 B2 B3 B4

Worklist



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B1 B2 B3 B4

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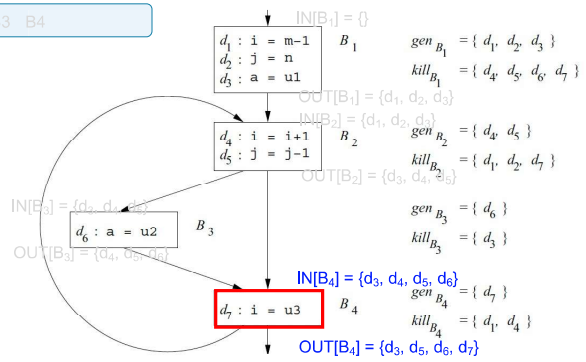

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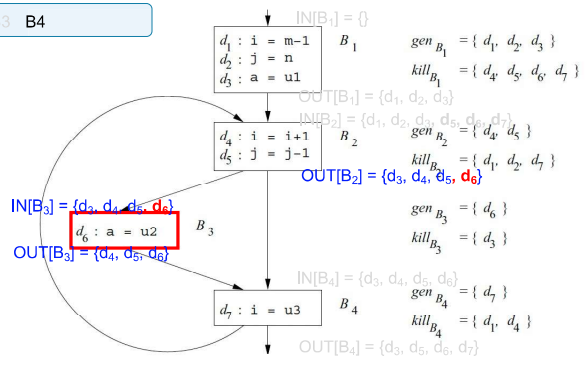

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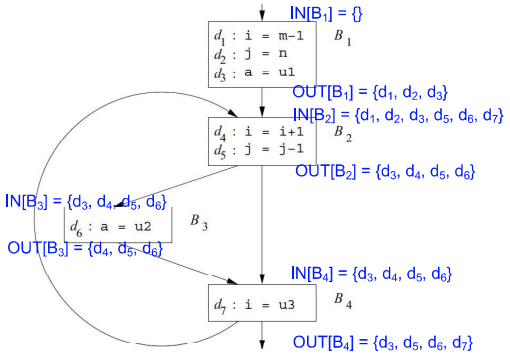
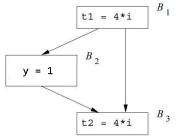
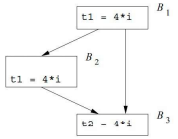


Available Expressions

- An expression x + y is available at a point p if
  - every path from the entry node to p evaluates x + y, and
  - after the last such evaluation prior to reaching p, there are no subsequent assignments to x or y.



4 * i is available before B_3



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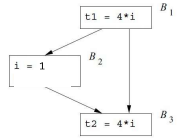

4 * i is NOT available before B_3

Available Expressions

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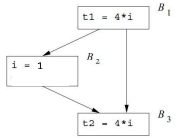
- The equations

$$\text{OUT}[B] = e\_gen_B \cup (\text{IN}[B] - e\_kill_B)$$

$$\text{IN}[B] = \bigcap_{P \text{ a predecessor of } B} \text{OUT}[P].$$

Why not ∪?



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Statement	Available Expressions
	$\emptyset$
a = b + c	
	$\{b+c\}$
b = a - d	
	$\{a-d\}$
c = b + c	
	$\{a-d\}$
d = a - d	
	$\emptyset$

Live Variable Analysis

- We wish to know if the value of a variable x at point p can be used along the path starting from p.

The values of y, z at this point are **live** as we can use them in the next statement.

The value of x at this point is **dead** because x will be redefined and the old value cannot be used any longer.

```
x = y + z
```


Live Variable Analysis

- **Application**: Register Allocation.

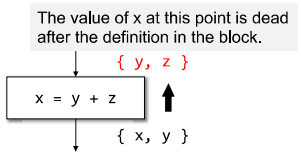
- A dead value
  - After a variable is revised, its previous value (in the register) is dead
  - We need to reload its value from the memory to the register

- A live value
  - A variable is not re-defined
  - After loading to a register, the value in the register is valid
  - No need to reload the value from the memory

Live Variable Analysis

- We wish to know if the value of a variable x at point p can be used along the path starting from p.
- **Backward data flow analysis.**
  - The same as forward analysis except for the direction.

$$\text{IN}[B] = use_B \cup (\text{OUT}[B] - def_B)$$
$$\text{OUT}[B] = \bigcup_{S \text{ a successor of } B} \text{IN}[S]$$



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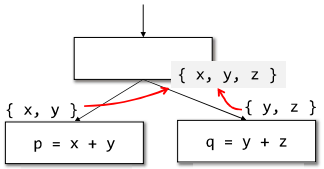
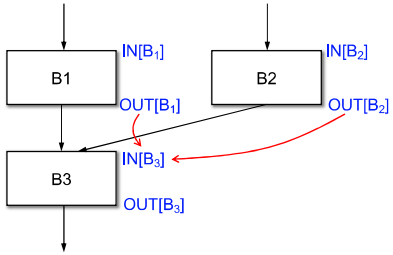


Summary

	Reaching Definitions	Live Variables	Available Expressions
Domain	Sets of definitions	Sets of variables	Sets of expressions
Direction	Forwards	Backwards	Forwards
Transfer function	$gen_B \cup (x - kill_B)$	$use_B \cup (x - def_B)$	$e\_gen_B \cup (x - e\_kill_B)$
Meet ($\wedge$)	$\cup$	$\cup$	$\cap$
Initialize	$\text{OUT}[B] = \emptyset$	$\text{IN}[B] = \emptyset$	$\text{OUT}[B] = U$

What's the Best Order of Blocks

- If we analyze B3 before B1 and B2, B3 may be analyzed again after B1 and B2 are analyzed

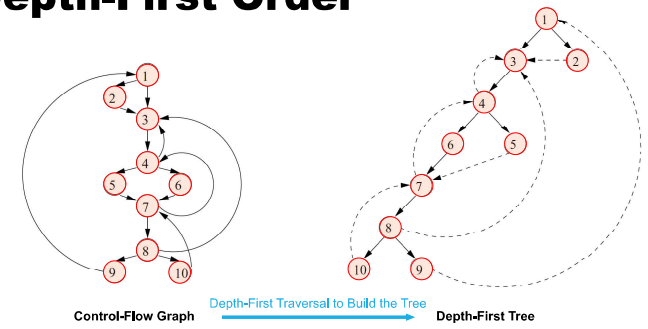


Question-I

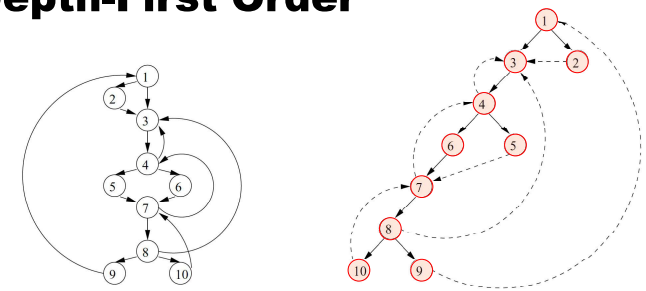
- **ForEach** basic block B: initialize IN[B] and OUT[B]; **EndFor**
- worklist = set of all basic blocks;
- **While** (!worklist.empty()) **Do**
- B = worklist.pop();
- IN[B] = $\wedge_{P \in preds(B)}$ OUT[P]; OUT[B] = f_B(IN[B])
- **If** OUT[B] changed: worklist.push(all B's successors); **EndIf**
- **EndWhile**

What's the best order of blocks?

Depth-First Order



Depth-First Order



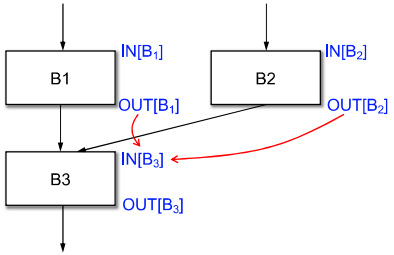
Depth-First Order is the Reversed Post-Order Sequence of the Depth-First Tree

Post-Order of the Tree: 10, 9, 8, 7, 6, 5, 4, 3, 2,1

Depth-First Order: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Depth-First Order

- It is exactly the topological order when the graph is acyclic
- It provides a topo-like order for cyclic graphs



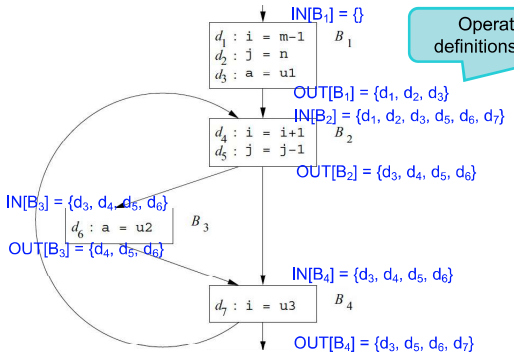
Question-II

- **ForEach** basic block B: initialize IN[B] and OUT[B]; **EndFor**
- worklist = set of all basic blocks;
- **While** (!worklist.empty()) **Do**
- B = worklist.pop();
- IN[B] = $\wedge_{P \in preds(B)}$ OUT[P]; OUT[B] = f_B(IN[B])
- **If** OUT[B] changed: worklist.push(all B's successors); **EndIf**
- **EndWhile**

Can the loop terminate?

Terminable if for each block B: IN[B] and OUT[B] is monotonically increased or decreased to a limit.

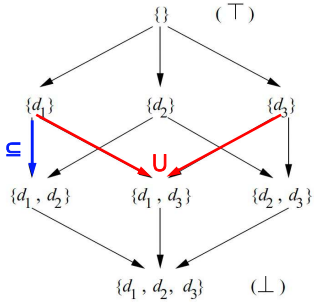
Recall: Reaching Definitions



Operations over subsets of definitions, forming a semi-lattice

Semi-Lattice

- A semilattice is a set V with a meet operation $\wedge$, such that
- (1) $x \wedge x = x$
- (2) $x \wedge y = y \wedge x$
- (3) $x \wedge (y \wedge z) = (x \wedge y) \wedge z$
- (4) $x \wedge \top = x$
- (5) $\bot \wedge x = \bot$

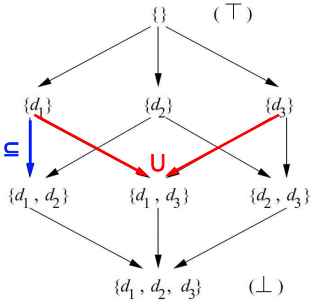


Semi-Lattice & Monotonicity

- Data Flow Equations
  - IN[B] = $\cup_{P \in preds(B)}$ OUT[P]
  - **OUT[B] = f_B(IN[B])**

Monotonicity:
- x ⊆ y
- => gen ∪ (x - kill) ⊆ gen ∪ (y - kill)
- => $f_B(x) \subseteq f_B(y)$

The semi-lattice has a finite height.



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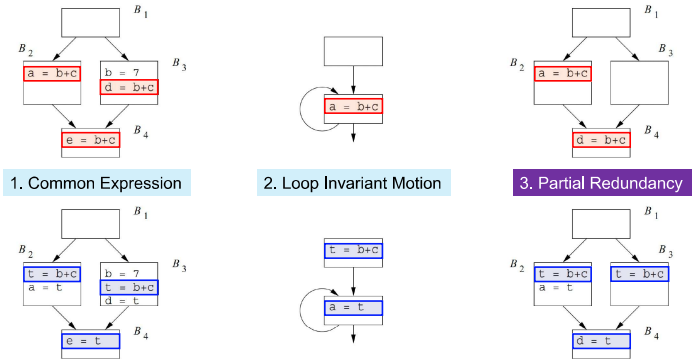
- **If** OUT[B] changed: worklist.push(all B's successors); **EndIf**
- **EndWhile**

If the semilattice of the framework is monotone and of finite height, then the algorithm is guaranteed to converge.

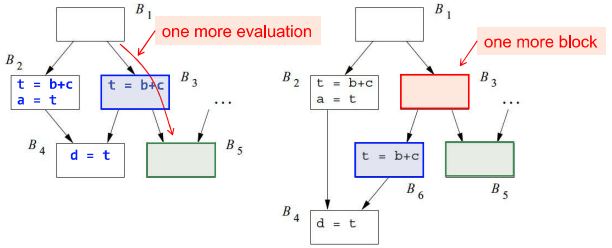
PART II: Partial Redundancy

Sources of Redundancy



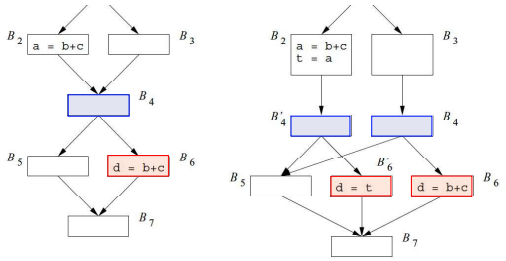
Eliminating All Redundancy?

- Not Possible!
- Unless we are allowed to create new blocks or duplicate code.



Eliminating All Redundancy?

- Not Possible!
- Unless we are allowed to create new blocks or duplicate code.



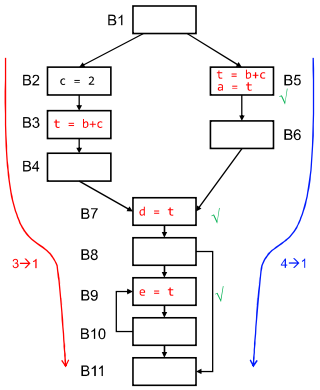
The Lazy Code Motion Problem

- All redundant computations of expressions that can be eliminated without code duplication are eliminated.
- The optimized program does not perform any computation that is not in the original program execution.
- Expressions are computed at the latest possible time.

The Lazy Code Motion Problem

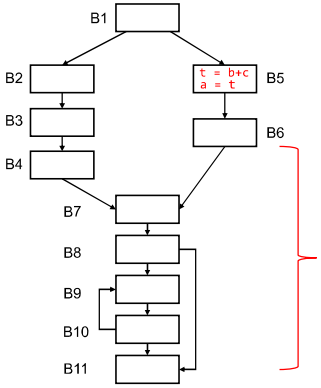
- **Step 1**: Anticipated Expressions (Backwards)
- **Step 2**: Available Expressions (Forwards)
- **Step 3**: Postponable Expressions (Forwards)
- **Step 4**: Used Expressions (Backwards)

Available Expressions (Forwards)



- Anticipated expr → Available expr
- An expression b+c is available at program point p if it is anticipated along all paths reaching p.
- Replace b+c with t.

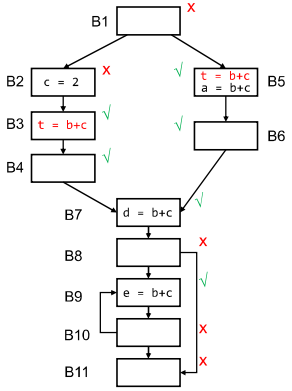
Used Expressions (Backwards)



- A simple backward data-flow pass to collect expressions used.
- Eliminate temporary variables that are used only once in the program.

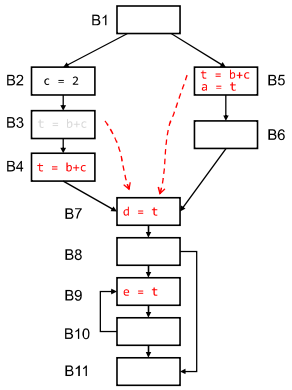
Assume the variable *t* is never used from B6 to B11?

Anticipated Expr (Backwards)



- An expression b+c is anticipated at a program point p if
  - all paths leading from point p eventually compute the expression b+c *from the values of b and c that are available at that point*

Postponable Expr (Forwards)



- If an expression is far from its use, what may happen?
- Why is t = b+c in B3 postponable?
- Why cannot t = b+c in B3 be post-poned to B7?

The Lazy Code Motion Problem

- **Step 1**: Anticipated Expressions (Backwards)
- **Step 2**: Available Expressions (Forwards)
- **Step 3**: Postponable Expressions (Forwards)
- **Step 4**: Used Expressions (Backwards)

Summary

	(a) Anticipated Expressions	(b) Available Expressions
Domain	Sets of expressions	Sets of expressions
Direction	Backwards	Forwards
Transfer function	$f_B(x) = e\_use_B \cup (x - e\_kill_B)$	$f_B(x) = (anticipated[B].in \cup x) - e\_kill_B$
Boundary	$\text{IN}[\text{EXIT}] = \emptyset$	
Meet ($\wedge$)	$\cap$	
Equations	$\text{IN}[B] = f_B(\text{OUT}[B])$ $\text{OUT}[B] = \bigwedge_{S,succ(B)} \text{IN}[S]$	
Initialization	$\text{IN}[B] = U$	

	(c) Postponable Expressions	(d) Used Expressions
Domain	Sets of expressions	Sets of expressions
Direction	Forwards	Backwards
Transfer function	$f_B(x) = (earliest[B] \sqcup x) - e\ use_B$	$f_B(x) = (e\ use_B \sqcup x) - latest[B]$
Boundary	$\text{OUT}[\text{ENTRY}] = \emptyset$	$\text{IN}[\text{EXIT}] = \emptyset$
Meet ($\wedge$)	$\cap$	$\cup$
Equations	$\text{OUT}[B] = f_B(\text{IN}[B])$ $\text{IN}[B] = \bigwedge_{P,pred(B)} \text{OUT}[P]$	$\text{IN}[B] = f_B(\text{OUT}[B])$ $\text{OUT}[B] = \bigwedge_{S,succ(B)} \text{IN}[S]$
Initialization	$\text{OUT}[B] = U$	$\text{IN}[B] = \emptyset$

Refer to Chapter 9, the Dragon book!

PART III: Constant Propagation

Constant Propagation

- Does a variable have a constant value at a program point?

- **Feature 1:** It has an unbounded set of possible data-flow values
- **Feature 2:** It is not distributive

Constant Propagation

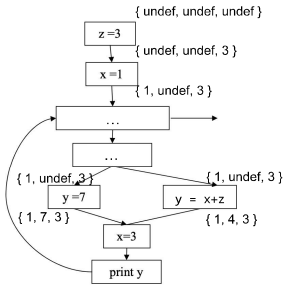
- Each variable v → UNDEF, Constant, NAC, denoted as m(v)

Transfer function of x = y + z

- m(y) = Constant, m(z) = Constant
- **=> m(x) = Constant**

- either of m(y) and m(z) is UNDEF
- **=> m(x) = UNDEF**

- otherwise
- **=> m(x) = NAC**

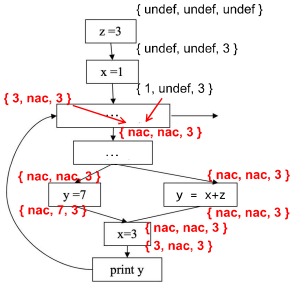


Constant Propagation

- Each variable v → UNDEF, Constant, NAC, denoted as m(v)

Merge Function: m(x) ∧ m(x')

- UNDEF ∧ Constant = Constant
- NAC ∧ Constant = NAC
- Constant ∧ Constant = Constant
- Constant ∧ Constant' = NAC



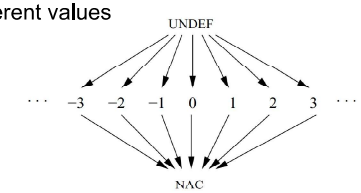
Constant Propagation

- Does a variable have a constant value at a program point?

- **Feature 1:** It has an unbounded set of possible data-flow values
- **Feature 2:** It is not distributive

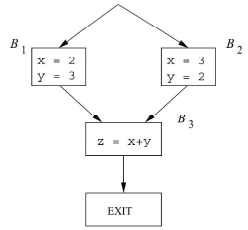
Monotonicity

- At each program point, m(x) always starts with UNDEF
  - ⇒become constant after assignment
  - ⇒become NAC if assigned different values
- At each program point
  - # NAC always increases



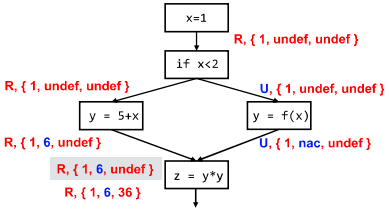
Non-distributivity

- f(a ∧ b) ≠ f(a) ∧ f(b) => Loss of precision in DFA!
- Meet-Then-Transfer ≠ Transfer-Then-Meet
- Meet-Then-Transfer
  - m(x) = NAC, m(y) = NAC **=> m(z) = NAC**
- Transfer-Then-Meet
  - m(z) = 5, m(z) = 5 **=> m(z) = 5**



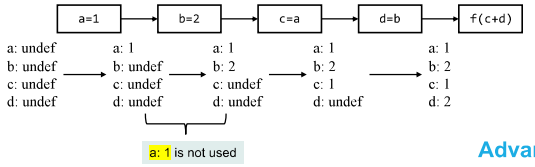
Conditional Const. Propagation

- Check the reachability of the code
- Check the branch condition => **Precision improvement!**

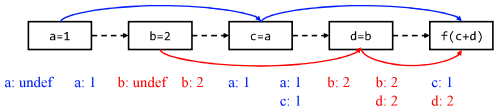


PART IV: Sparse DFA

Constant Propagation



Advantages?



Sparse DFA

- **Sparse** Dataflow Analysis Relations to SSA?
  - **Temporal sparsity:** Skip unnecessary control flows
  - **Space sparsity:** Preserve necessary dataflow facts at a program point